

Nurture Course COURSE

PHASE : SPEED & ACCURACY TEST

TARGET : PRE-NURTURE 2023

Test Type : PRACTICE TEST

Test Pattern : NEET (UG)

TEST DATE : _____

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	3	3	4	1	4	3	3	4	4	1	3	2	2	4	1	3	1	1	4	4	3	4	4	2	1	4	1	3	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45															
A.	3	2	1	1	1	2	3	3	1	3	3	3	2	4	1															



HINT - SHEET

1. **Ans (2)**

$$R_H = \frac{Q}{t} = e_r \sigma A (T^4 - T_0^4)$$

$$\frac{R_{H_1}}{R_{H_2}} = \frac{T_1^4 - T_0^4}{T_2^4 - T_0^4}$$

$$\frac{45}{R_{H_2}} = \frac{(800)^4 - (400)^4}{(1200)^4 - (400)^4} = \frac{(400)^4 [2^4 - 1]}{(400)^4 [3^4 - 1]} = \frac{15}{80}$$

$$R_{H_2} = \frac{45 \times 80}{15} = 240 \text{ watt}$$

2. **Ans (3)**

$$\Delta U = \frac{f}{2} \mu R (T_2 - T_1) = \frac{f}{2} (P_2 V_2 - P_1 V_1)$$

$$\Delta U = \frac{3}{2} (4P_0 V_0 - P_0 V_0) = 4.5 P_0 V_0$$

3. **Ans (3)**

$$W_{ABC} = W_{AB} + W_{BC}$$

$$= \frac{1}{2} (2P_0 + P_0) (V_0 - 2V_0) + \frac{1}{2} (3P_0 + 2P_0) (3V_0 - V_0)$$

$$= -\frac{3P_0}{2} + 5P_0 V_0 = \frac{7P_0 V_0}{2}$$

4. **Ans (4)**

$$W_{net} = W_{AB} + W_{BC} + W_{CD} + W_{DA}$$

$$= 2P(2V - V) + R(2T) \ln \frac{2P}{P} + P(2V - 4V) + RT \ln \frac{P}{2P}$$

$$= 2PV + 2RT \ln 2 - 2PV - RT \ln 2 = RT \ln 2$$

5. **Ans (1)**

$$TV^{n-1} = \text{constant}$$

$$TV^2 = \text{constant}$$

$$n-1 = 2$$

$$n = 3$$

$$C = C_V + \frac{R}{1-n} = \frac{3R}{2} + \frac{R}{1-3} = \frac{3R}{2} - \frac{R}{2} = R$$

6. **Ans (4)**

$$PT^{-\alpha} = \text{constant} \quad \dots (i)$$

$$P^{1-\gamma} T^\gamma = \text{constant}$$

$$P T^{\gamma/(1-\gamma)} = \text{constant} \quad \dots (ii)$$

Compare equation (i) and (ii)

$$\frac{\gamma}{1-\gamma} = -\alpha$$

$$\frac{\frac{7}{5}}{1 - \frac{7}{5}} = -\alpha$$

$$\alpha = 3.5$$

7. **Ans (3)**

$$W_1 = +ve, W_2 = 0, W_3 = -ve$$

$$FLOT = Q = W + \Delta U$$

DU same for all processes

$$\text{So, } Q_1 > Q_2 > Q_3$$

8. **Ans (3)**

$$W_{net} = W_{ABC} + W_{ADE}$$

$$= \frac{1}{2} (2-1) \times (4-2) - \frac{1}{2} (3-2) (2-1)$$

$$= 1 - \frac{1}{2} = \frac{1}{2} \text{ J}$$

9. **Ans (4)**

$$(C_P)_{\min} = \frac{\mu_1 C_{P_1} + \mu_2 C_{P_2} + \mu_3 C_{P_3}}{\mu_1 + \mu_2 + \mu_3}$$

$$= \frac{4 \times 7 \frac{R}{2} + 2 \times \frac{5R}{2} + 1 \times \frac{8R}{2}}{4 + 2 + 1} = \frac{23R}{7}$$

10. **Ans (4)**

$$P_0 = \frac{1}{3} \frac{M}{V} \cdot v_{\text{rms}}^2, P' = \frac{1}{3} \times \frac{2M}{V} \times \left(\frac{v_{\text{rms}}}{2} \right)^2 = \frac{P_0}{2}$$

11. **Ans (1)**

$$\frac{\Delta Q}{W} = \frac{\mu C_P \Delta T}{\mu R \Delta T} = \frac{C_P}{R} = \frac{\frac{5}{2}R}{R} = 2.5$$

12. **Ans (3)**

$$\frac{X - (-125)}{500} = \frac{Y - (-70)}{40}$$

For $Y = 50$

$$X = 1375.0^\circ\text{X}$$

13. **Ans (2)**

the change in length of rod due to increase in temperature in absence of walls is

$$\Delta \ell = \ell \alpha \Delta T = 1000 \times 10^{-4} \times 20 \text{ mm} = 2 \text{ mm}$$

But the rod can expand upto 1001 mm only.

At that temperature its natural length is = 1002 mm.

$$\therefore \text{compression} = 1 \text{ mm}$$

$\therefore$ mechanical stress

$$= Y \frac{\Delta \ell}{\ell} = 10^{11} \times \frac{1}{1000} = 10^8 \text{ N/m}^2$$

14. **Ans (2)**

0% of K.E = Heat

$$\frac{1}{2} \times \left(\frac{1}{2} m v^2 \right) = m s \Delta T + m L$$

$$\Rightarrow \frac{v^2}{4} = S \Delta T + L$$

$$\Rightarrow \frac{v^2}{4} = 125 (327 - 27) + 2.5 \times 10^4$$

$$\Rightarrow v = 500 \text{ m/s}$$

15. **Ans (4)**

$$T_{\text{mix}} = \frac{M_1 S_1 T_1 + M_2 S_2 T_2}{M_1 S_1 + M_2 S_2}$$

$$32 = \frac{m_1 \times 0.2 \times 40 + 100 \times 0.5 \times 20}{m_1 \times 0.2 + 100 \times 0.5}$$

$$m_1 = 375 \text{ g}$$

16. **Ans (1)**

$$\frac{\rho_1}{\rho_2} = \frac{3}{4}, \frac{S_1}{S_2} = \frac{4}{3}$$

$$\frac{H_{C_1}}{H_{C_2}} = \frac{M_1 S_1}{M_2 S_2} = \frac{V \rho_1 S_1}{V \rho_2 S_2}$$

$$\Rightarrow \frac{H_{C_1}}{H_{C_2}} = \frac{3}{4} \times \frac{4}{3} = 1 : 1$$

17. **Ans (3)**

Heat released by steam in conversion to water at 100°C is $Q_1 = mL$

$$= 4 \times 540 = 2160 \text{ cal.}$$

Heat required to raise temperature of water from 46°C to 100°C is $Q_2 = m S \Delta Q$

$$= 20 \times 1 \times 54 = 1080 \text{ J}$$

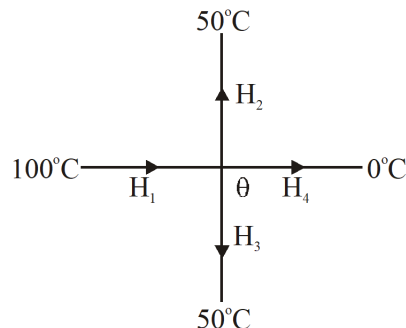
$$= Q_1 > Q_2 \text{ and } \frac{Q_1}{Q_2} = 2$$

Hence all steam is not converted to water only half steam shall be converted to water.

$$\text{Final mass of water} = 20 + 2 = 22 \text{ gm}$$

18. **Ans (1)**

Let θ be the temperature of junction. H_1, H_2, H_3 and H_4 be the currents. Then,



$$H_1 = H_2 + H_3 + H_4$$

$$\text{or } \frac{100 - \theta}{R} = \frac{\theta - 50}{R} + \frac{\theta - 50}{R} + \frac{\theta - 0}{R}$$

(R = Thermal resistance of each part of two rods Solving these two equations, we get)

$$\theta = 50^\circ\text{C}$$

19. **Ans (1)**

Rate of heat transfer is same through all walls

$$\frac{K_1 \cdot A \cdot (10)}{40 \text{ cm}} = \frac{K_2 A (20)}{20 \text{ cm}} = \frac{K_3 A (40)}{10 \text{ cm}}$$

$$\Rightarrow \frac{K_1}{4} = K_2 = 4K_3$$

$$\Rightarrow K_1 = 4K_2 = 16K_3$$

$$\Rightarrow K_1 \text{ is greatest}$$

20. Ans (4)

$$t = \frac{\rho L}{2K\theta} (x_2^2 - x_1^2) \Rightarrow t \propto (x_2^2 - x_1^2)$$

$$\Rightarrow \frac{t}{t'} = \frac{(x_2^2 - x_1^2)}{(x_2'^2 - x_1'^2)}$$

$$\Rightarrow \frac{7}{t'} = \frac{(1^2 - 0^2)}{(2^2 - 1^2)} \Rightarrow t' = 21 \text{ hours}$$

21. Ans (4)

$$\frac{Q}{At} = \frac{K(\theta_1 - \theta_2)}{d} = \text{constant}$$

$$\theta_1 - \theta_2 = 20^\circ\text{C}$$

$$\therefore K_A \left(\frac{\theta_1 - \theta}{d} \right) = K_B \left(\frac{\theta - \theta_2}{d} \right)$$

$$3K_B(\theta_1 - \theta) = K_B(\theta - \theta_1 + 20)$$

$$4(\theta_1 - \theta) = 20$$

$$\theta_1 - \theta = 5^\circ\text{C}$$

22. Ans (3)

$$\frac{dQ}{dt} = e\sigma AT^4$$

$$\Rightarrow \frac{dQ}{dt} \propto r^2 T^4$$

$$\Rightarrow \frac{(dQ/dt)_1}{(dQ/dt)_2} = \frac{(1)^2 \cdot (4000)^4}{(4)^2 (2000)^4} = \frac{1}{1}$$

$$\Rightarrow \left(\frac{dQ}{dt} \right)_1 = \left(\frac{dQ}{dt} \right)_2$$

24. Ans (4)

$$R_C \propto \frac{A}{V} \text{ where } A = 4\pi r^2 \text{ and } V = \frac{4}{3}\pi r^3$$

$$\Rightarrow R_C \propto \frac{1}{r}$$

25. Ans (2)

$$\text{Given } \frac{e_A}{e_B} = \frac{1}{e} \therefore \frac{dQ}{dt} = e\sigma AT^4$$

$$\frac{e_A}{e_B} = \left(\frac{T_B}{T_A} \right)^4 \Rightarrow \frac{(\lambda_m)_A}{(\lambda_m)_B} = \left(\frac{1}{e} \right)^{1/4}$$

26. Ans (1)

$$R_C \propto (T^4 - T_0^4)$$

$$\frac{H}{R_C} = \frac{600^4 - 300^4}{900^4 - 300^4}$$

$$R_C = \frac{16H}{3}$$

27. Ans (4)

$$E = \sigma T^4 \Rightarrow R = \sigma \left(\frac{b}{\lambda_m} \right)^4$$

$$\Rightarrow \left(\frac{R}{\sigma} \right)^{1/4} = \frac{b}{\lambda_m}$$

$$\Rightarrow \lambda_m = b \left(\frac{\sigma}{R} \right)^{1/4}$$

28. Ans (1)

$$\text{Here, } T = V \tan 45^\circ + T_0$$

$$\Rightarrow T = V + T_0$$

$$\text{And } P = \frac{nRT}{V}$$

$$\therefore \frac{PV}{R} = V + T_0 \text{ (Since, } n=1)$$

$$(P-R)V = RT_0$$

Therefore, graph will be rectangular hyperbola.

29. Ans (3)

The kinetic energy of gas w.r.t. centre of mass of the system K.E. = $\frac{5}{2} nRT$

Kinetic energy of gas w.r.t. ground = Kinetic energy of centre of mass w.r.t. ground + Kinetic energy of gas w.r.t. centre of mass

$$\text{K.E.} = \frac{1}{2} MV^2 + \frac{5}{2} nRT$$

30. Ans (2)

Mass of $H_2 = M_1 g$

Mass of $He = M_2 g$

$$M_1 + M_2 = 20 \text{(i)}$$

$$PV = \mu_{\text{mix}} RT$$

$$P \left(\frac{M}{\rho} \right) = \left(\frac{M_1}{M_{W_1}} + \frac{M_2}{M_{W_2}} \right) RT$$

$$\frac{10^6 \times 20 \times 10^{-3}}{1} = \left(\frac{M_1}{2} + \frac{M_2}{4} \right) \times 8.31 \times 300$$

$$2M_1 + M_2 = 32 \text{(ii)}$$

By eq. (i) and (ii)

$$\Rightarrow M_1 = 12 \text{ g and } M_2 = 8 \text{ g}$$

So $(\mu)_{H_2} = 6 \text{ mole and } (\mu)_{He} = 2 \text{ mole}$

31. **Ans (3)**

$$\text{For A} \Rightarrow V_{\text{rms}}^2 A = 3V_x^2 = 3\omega^2$$

$$\text{For B} \Rightarrow V_{\text{rms}}^2 B = V^2$$

$$V_{\text{rms}}^2 \propto \frac{1}{m}, \text{ So, } \frac{3\omega^2}{v^2} = \frac{2m}{m} \Rightarrow \frac{\omega^2}{v^2} = \frac{2}{3}$$

32. **Ans (2)**

$$U = a + b PV = a + b nRT$$

$$C_V = \frac{1}{n} \frac{dU}{dT} = bR \Rightarrow C_P = C_V + R = (b+1)R$$

$$\text{Now } \frac{C_P}{C_V} = \frac{b+1}{b}$$

33. **Ans (1)**

Use the relation

$$V_{\text{rms}} = \sqrt{\frac{V_1^2 + V_2^2 + V_3^2 + V_4^2}{4}}$$

and get the answer $5\sqrt{3}V$

34. **Ans (1)**

$$E = \frac{\mu fRT}{2}$$

$$E_{\text{He}} = (2) \left(\frac{3}{2} RT \right)$$

$$E_{\text{N}_2} = 4 \left(\frac{5}{2} RT \right)$$

$$E_{\text{NH}_3} = 5 \left(\frac{6}{2} RT \right)$$

$$E_{\text{Total}} = 28RT$$

35. **Ans (1)**

$$V \propto \sqrt{\frac{T}{m_0}}$$

$$\frac{V_{03}}{V_0} = \left(\sqrt{\frac{m_0}{m_{03}}} \right) \left(\sqrt{\frac{T_{03}}{T_0}} \right)$$

$$\frac{V_{03}}{V_0} = \sqrt{\frac{16}{48} \times \frac{1}{3}} = \frac{1}{3}$$

$$\Rightarrow V_0 = 3V$$

36. **Ans (2)**

$$\text{For an adiabatic process } P_1^{1-\gamma} T_1^\gamma = P_2^{1-\gamma} T_2^\gamma$$

$$1^{1-\gamma} 100^\gamma = 16^{1-\gamma} 200^\gamma$$

$$\Rightarrow 16^{1-\gamma} = \left(\frac{1}{2} \right)^\gamma \text{ or } \left(\frac{1}{16} \right)^{\gamma-1} = \left(\frac{1}{2} \right)^\gamma$$

$$\left(\frac{1}{2} \right)^{4\gamma-4} = \left(\frac{1}{2} \right)^\gamma \Rightarrow 4\gamma-4 = \gamma \Rightarrow \gamma = \frac{4}{3}$$

37. **Ans (3)**

$$dQ = nC_P dT$$

$$C_P = \frac{dQ}{ndT} = \frac{50R}{4 \times 5} = \frac{5}{2}R \Rightarrow \text{monoatomic}$$

38. **Ans (3)**

$$W_{AB} = \frac{1}{2} (2P_0 - P_0) (2V_0 - V_0) = 1.5 P_0 V_0$$

$$P_0 V_0 = nRT_A \text{ \& } 2P_0 2V_0 = nRT_B$$

$$\Rightarrow T_B = 4T_A, \Delta T = 3T_A$$

$$\Delta U_{AB} = nC_V \Delta T = nR \frac{3}{2} (3T_A)$$

$$\Delta U = \frac{9}{2} nRT_A = \frac{9}{2} P_0 V_0$$

$$Q = \Delta U + W = 6P_0 V_0$$

39. **Ans (1)**

$$\rho = \frac{PM}{RT} \text{ and } \rho \propto \frac{1}{V}$$

during AB, r is constant and hence volume there for work done is zero.

During BC $P \propto r$

i.e. $T = \text{constant}$ and hence U is constant

40. **Ans (3)**

Let initial pressure of three samples are

$$P_x, P_y, P_z \text{ then } P_x V^{5/3} = P(2V)^{5/3}$$

$$P_y = P$$

$$P_z V = P(2V)$$

$$P_x : P_y : P_z = 2\sqrt[3]{4} : 1 : 2$$

41. **Ans (3)**

$$W_{\text{clockwise cycle}} = + (\text{Area 12341})$$

$$= (6V_0 - 3V_0) (5P_0 - 3P_0) = 6P_0 V_0$$

42. **Ans (3)**

$$\Rightarrow T^2 \propto V^3$$

$$\left(\frac{PV}{nR} \right)^2 \propto V^3$$

$$P^2 V^{-1} = \text{const}$$

$$PV^{-1/2} = \text{const} \Rightarrow x = -1/2$$

$$C = C_V + \frac{R}{1-x} = \frac{5R}{2} + \frac{2R}{3} = \frac{19R}{6}$$

43. Ans (2)

$$\text{Slope } \frac{Q}{\Delta T} = \frac{2500}{100}$$

$$\frac{nC_p\Delta T}{\Delta T} = 25 \Rightarrow C_p = 25$$

$$\Rightarrow \frac{\gamma R}{\gamma - 1} = 25 \Rightarrow \gamma = 3(\gamma - 1) \Rightarrow 3 = 2\gamma$$

$$\Rightarrow \gamma = 3/2$$

44. Ans (4)

$$\text{Isobaric} \Rightarrow Q = nC_p\Delta T$$

$$Q = 1 \times \frac{5}{2}R \times 10 \dots (1)$$

$$Q_2 = nC_p\Delta T$$

$$Q_2 = 1 \times \frac{7R}{2} \times 5 \dots (2)$$

$$Q_2 = \frac{7}{10}Q$$

45. Ans (1)

$$W_1 = P(V_2 - V_1) = P(2V_0 - V_0) = PV_0$$

$$W_2 = nRT \ln \frac{V_2}{V_1}$$

$$= PV_0 \ln \frac{2V_0}{V_0} = PV_0 \ln(2)$$

$$\Rightarrow W_2 = W_1 \ln(2)$$